Course Scheduling and Classroom Allocation Database for a University

VBA 215E Term Project – Topic Proposal | Sector: Education | Group 03

070230106 Cansu Ergöz 070230044 Sıla Başaran 070250501 Ali Bulut 070240517 Didem Uzun 070250503 Melih Okumuş

1. Problem Definition

In a large university with many faculties, the weekly timetable is usually built in pieces: each department prepares its own sections and rooms are then matched on spreadsheets passed between departments and the planning office. Because these files are not linked, a room can be given to two sections at the same hour, an instructor can end up in two places at once, or a 120-student course is put in a 60-seat room. Changes made during the term are hard to trace, and nobody can easily say how much of the classroom capacity is really used.

We will design a database for the **Academic Planning Office** of a university modelled on ITU: several faculties, each with its own buildings, plus one central lecture-hall building that does not belong to any faculty and is used for common (pool) courses such as Mathematics, Physics and English. The database will keep rooms, courses, sections and weekly sessions together, block clashes and unsuitable rooms through constraints, and report on room usage.

2. Users

- **Planning office staff:** place sections into rooms and slots, decide on room requests, follow changes.
- **Department secretaries:** enter courses and each term's sections (quota, instructor, room needs).
- **Instructors:** view their timetable, request rooms for make-up classes, seminars or club events.
- **Deans:** use utilization and peak-hour reports for space planning.

3. Scope

Included: faculties, departments, courses and prerequisites, buildings, rooms and their equipment, terms, sections and instructors, weekly session–room–time assignments, clash and suitability checks, room reservations with approval, a change log and reports, plus the data warehouse and MongoDB parts.

Not included: automatic timetable generation, student registration and grades (we only keep the number of enrolled students), exam scheduling, maintenance and any user interface.

4. Main Entities

Faculty / Department	A faculty has departments; each department belongs to one faculty and owns courses and instructors.	**Equipment**	Projector, smart board, etc. Linked to rooms through a bridge table that holds the quantity.
Course	Catalog course, either regular or pool. Prerequisites are stored as a recursive relationship on Course.	**Term**	Academic term with start and end dates, e.g. 2026-2027 Fall.
Building	Belongs to one faculty, or to none (the central building).	**Section**	A course opened in a term (weak entity on Course + Term), with quota and enrolled count.
Room	Weak entity of Building. Supertype of Classroom, Computer Lab and Amphitheater.	**Class Session**	One weekly block of a section: day, start slot, length, room.
Instructor	Academic staff with a title, in one department. Linked to sections through a bridge table.	**Reservation**	One-off room request with date, time, purpose and status; planning staff decide on it.

5. Example Business Questions

1. Which rooms are used the most and the least during the week, and which sections fill less than half of the room they are in?
2. Which days and hours are the busiest, and when are most rooms empty?
3. How many hours does each instructor teach per week, and who is close to the 20-hour limit?
4. Which computer labs (or rooms of a given type and size) are free on a given day and hour?
5. How does utilization compare between faculties and the central building?
6. Which rooms receive the most reservation requests, and what share of requests are rejected?

6. Initial Business Rules

Organization and courses

1. Each department belongs to exactly one faculty. Each instructor works in one department and has an academic title.
2. Each course is owned by one department and is either a regular course or a pool course offered to all faculties.
3. A course may have several prerequisite courses, but it cannot be its own prerequisite.

Buildings and rooms

4. A building belongs to one faculty or to none; the building with no faculty is the central lecture-hall building.
5. Room numbers repeat between buildings, so a room is identified by its building and room number together.
6. Every room is a classroom, a computer lab or an amphitheater. Labs also store the number of computers, amphitheaters the sound system.
7. Room capacity must be greater than zero. A room can have many equipment types, each with a quantity.

Terms, sections and sessions

8. A course can be opened as one or more sections in a term (Section 1, 2, …). The enrolled count can never exceed the quota.
9. A section can have more than one instructor, but exactly one is responsible. A section may require a room type and equipment.
10. Classes run Monday to Friday in eight one-hour slots (08:30–12:30 and 13:30–17:30), i.e. 40 slots per room per week. Room utilization = used slots / 40.
11. A session takes 2 or 3 consecutive slots in a single room and cannot run over the lunch break. A section can have several sessions a week, but they cannot overlap.

Assignment checks

12. A room cannot have two sessions in the same slot, and an instructor cannot teach two sessions in the same slot.
13. The room's capacity must be at least the section quota, and the room must have the type and equipment the section needs.
14. Regular courses go to buildings of their own faculty; pool courses go to the central building.
15. An instructor cannot teach more than 20 hours per week.

Reservations and tracking

16. Instructors and staff can request any room for a one-off event on a date, with the same slot rules. A request stays Pending until planning staff approve or reject it; it can also be cancelled.
17. An approved reservation cannot overlap another approved one, or a regular session in the same room on that weekday of the term.
18. Each change to a session's room or time is logged (old/new value, who, when, why).

Not decided yet: whether two sessions of one section on the same day need an empty slot between them (to be settled before Week 9).